BRAKE CHOPPER — TWO COMMAND SOURCES, ONE GATE DRIVE

FIRMWARE / HARDWARE DIODE OR

BRAKE_PWM

R502
33R

D502
BAT54CM

BRAKE_PWM_DRV

R501
10k

GND

R505
10k

GND

LOCAL GATE DRIVE / POWER SWITCH

VCC

C501
1uF 25V

C502
100nF 50V

GND

U501
UCC27517ADBVR

VDD
IN+
IN−
GND
OUT

GND

R503
4.7R

TP501
BRAKE_GATE

Q501
ISZ080N10NM6 100V

D501
BZT52C12 12V

R504
10k

GND

VM

J501

EXTERNAL
10R / 250W*

BRAKE RESISTOR

TP502
BRAKE_SWITCH

Short / twisted resistor leads.

Validate Q501 drain overshoot.

Q501: D = pads 5–8; S = pads 1–3; G = pad 4.

CAPACITOR UN-EXPLODER

Board-bus overvoltage backup — NOT a BMS or battery protector.

VM

R506
470k 1% 75V

R510
1M 1%

3V3

C503
100nF 50V

GND

U502
TLV3012BIDCKR

VDD
IN+
IN−
OUT
REF
GND

R509
13.3k 1%

C504
1nF C0G

GND

BRAKE_OV_CMD

The positive-feedback resistor is wired back to IN+.

Internal 1.242 V reference returns directly to IN−.

U502: nominal ON 45.8 V / OFF 44.1 V. Board-bus backup only; thresholds have tolerance and dynamic overshoot.

*Provisional external resistor: RH25010R00FE01, 10R / 250W WITH heatsink. About 212W at 46V; size for stopping energy.

Place U501 decoupling / R503 / D501 at Q501. Connect U501 pin 2 (GND) and pin 4 (IN−) to GND; Kelvin return to Q501 source.

Brake NTC removed. Brake PWM and hardware bus-overvoltage clamp remain active.

5V PWM FAN / SHARED PROTECTED SUPPLY

EXT_VCC

C505
100nF

C506
4.7uF

J502
5V PWM FAN

GND
+5V
TACH
PWM
MOUNT

R511
100R

FAN_PWM

FAN_PWM_5V

U503
74AHCT1G125W5-7

VCC
A
Y
OE
GND

R512
100k

U604 on External feedback supplies J601 + J502: 520mA nominal shared limit.
PC1 / LPTIM1_OUT: 25kHz, non-inverting. R512 holds the buffer input LOW at reset.
Firmware: keep PWM LOW on USB-only power; allow running only when motor VM is present.
0% stops the specified Noctua fan. Two-wire fans on pins 1/2 run whenever 5V_SYS is powered.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /Brake chopper/
File: 07_Brake_Chopper.kicad_sch

Title: Brake chopper / overvoltage protection

Size: A3	Date: 2026-09-16	**Rev: A1-Drawing**
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EXTERNAL HALL / ABI: 3.3V or 5V logic -> 3.3V Schmitt buffer -> reset isolation

Joint feedback supply: fixed, protected 5V shared with J502 fan.
J601.1 = 5V output; J601.2/6 = GND. No power input.
For a 5V Hall/ABI encoder; no voltage selector.
Sensor + pull-ups target <=25mA; shared accessory target <=300mA.

D601 rail isolated from supplies.
C604 is required; no 3V3 clamp tie.

INPUT OPTIONS
Default: 3.3V / 5V push-pull; pull-ups DNP.
Open-drain: fit R608-R610 (4.7k to protected 5V).
Do not fit pull-ups for externally driven push-pull.
100k pull-downs define disconnected inputs.
Single-ended only; common ground required.

D601
TPD3E001DRYR
NC
GND
VCC
IO3
IO2
IO1
FB_ESD_RAIL
C604
100nF 50V

J601
EXT ENCODER / HALL
MP
MOUNT

R608
4.7k DNP
R605
100R
R615
100k
R609
4.7k DNP
R606
100R
R616
100k
R610
4.7k DNP
R607
100R
R617
100k

U603A
U603C
U603B
74LVC3G17

U602A
TMUX1574PW
TP1015
FEEDBACK_A_H1
TP1016
FEEDBACK_B_H2
FEEDBACK_I_H3

U604
TPS2553DBVR
5V_SYS
C606
4.7uF
R618
49.9k 1%
R604
10k
EXT_POWER_FAULT_N

U604: R618=49.9k 1% =>520mA nominal shared limit.
EXT_VCC feeds J601 sensor + J502 fan + U503 PWM buffer.
EXT_POWER_FAULT_N -> PC13; shared branch fault.
Stop torque on fault or external-feedback loss; verify fan startup margin.

C607
100nF 16V X7R

U602B
TMUX1574PW
VBUS_MUX_IN
VBUS_SENSE

U603D
74LVC3G17
C605
100nF

R305
10k
R306
10k
R307
10k
C603
100nF
INPUT_MUX_ENABLE_N
EXT_FEEDBACK_SELECT
U602C
TMUX1574PW

Reset: mux disabled, PB8/BOOT0 held low by R307.
EN_N=0 enables VM sensing; wait >=2ms before ADC read.
SEL=0 grounds TIM4 inputs; SEL=1 connects external Hall/ABI.
Onboard absolute-angle SPI is independent of this switch.

3.3V / 5V external inputs; interface revision
ShiroFOC
Sheet: /External feedback/
File: 09_Feedback_Selection.kicad_sch
Title: External Hall / ABI conditioning and reset isolation
Size: A3 | Date: 2026-09-26 | **Rev: A1-Drawing**
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